

```
In [2]: import pandas as pd
```

```
In [3]: pd.__version__
```

```
Out[3]: '2.2.3'
```

```
In [4]: emp = pd.read_excel(r"C:\Users\chamb\Downloads\Rawdata.xlsx")
emp
```

```
Out[4]:
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience#\$	34 years	Mumbai	5^00#0	2+
1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderbad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^\$0	10+

```
In [5]: id(emp)
```

```
Out[5]: 2254073015200
```

```
In [6]: emp.shape
```

```
Out[6]: (6, 6)
```

```
In [7]: emp.columns
```

```
Out[7]: Index(['Name', 'Domain', 'Age', 'Location', 'Salary', 'Exp'], dtype='object')
```

```
In [8]: emp.head
```

```
Out[8]: <bound method NDFrame.head of
```

	Name	Domain	Age	Location
0	Mike	Datascience#\$	34 years	Mumbai
1	Teddy^	Testing	45' yr	Bangalore
2	Uma#r	Dataanalyst^^#	NaN	NaN
3	Jane	Ana^^lytics	NaN	Hyderbad
4	Uttam*	Statistics	67-yr	NaN
5	Kim	NLP	55yr	Delhi

```
In [9]: emp.tail
```

```
Out[9]: <bound method NDFrame.tail of
```

	Name	Domain	Age	Location
0	Mike	Datascience#\$	34 years	Mumbai
1	Teddy^	Testing	45' yr	Bangalore
2	Uma#r	Dataanalyst^^#	NaN	NaN
3	Jane	Ana^^lytics	NaN	Hyderbad
4	Uttam*	Statistics	67-yr	NaN
5	Kim	NLP	55yr	Delhi

```
In [10]: emp.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   Name        6 non-null      object
 1   Domain      6 non-null      object
 2   Age         4 non-null      object
 3   Location    4 non-null      object
 4   Salary      6 non-null      object
 5   Exp         5 non-null      object
dtypes: object(6)
memory usage: 420.0+ bytes
```

```
In [11]: emp.isnull()
```

```
Out[11]:
```

	Name	Domain	Age	Location	Salary	Exp
--	------	--------	-----	----------	--------	-----

0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	True	True	False	False
3	False	False	True	False	False	True
4	False	False	False	True	False	False
5	False	False	False	False	False	False

```
In [12]: emp.isna()
```

```
Out[12]:
```

	Name	Domain	Age	Location	Salary	Exp
--	------	--------	-----	----------	--------	-----

0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	True	True	False	False
3	False	False	True	False	False	True
4	False	False	False	True	False	False
5	False	False	False	False	False	False

```
In [13]: emp.isnull().sum()
```

```
Out[13]: Name      0
Domain    0
Age       2
Location  2
Salary    0
Exp       1
dtype: int64
```

```
In [14]: emp['Name']
```

```
Out[14]: 0      Mike
         1      Teddy^
         2      Uma#r
         3      Jane
         4      Uttam*
         5      Kim
         Name: Name, dtype: object
```

Data Cleaning

```
In [15]: emp
```

```
Out[15]:
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience#\$	34 years	Mumbai	5^00#0	2+
1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderbad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^\$0	10+

```
In [16]: emp['Name']
```

```
Out[16]: 0      Mike
         1      Teddy^
         2      Uma#r
         3      Jane
         4      Uttam*
         5      Kim
         Name: Name, dtype: object
```

```
In [17]: emp['Name'] = emp['Name'].str.replace(r'\W', '', regex=True) #\W= non word chara
```

```
In [18]: emp['Name']
```

```
Out[18]: 0      Mike
         1      Teddy
         2      Umar
         3      Jane
         4      Uttam
         5      Kim
         Name: Name, dtype: object
```

```
In [19]: emp
```

Out[19]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience#\$	34 years	Mumbai	5^00#0	2+
1	Teddy	Testing	45' yr	Bangalore	10%%000	<3
2	Umar	Dataanalyst^^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderbad	2000^0	NaN
4	Uttam	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^\$0	10+

In [20]: `emp['Domain']`

Out[20]:

```
0    Datascience#$
1         Testing
2    Dataanalyst^^#
3         Ana^^lytics
4         Statistics
5             NLP
Name: Domain, dtype: object
```

In [21]: `emp['Domain'] = emp['Domain'].str.replace(r'\W', '', regex=True)`

In [22]: `emp['Domain']`

Out[22]:

```
0    Datascience
1         Testing
2    Dataanalyst
3         Analytics
4         Statistics
5             NLP
Name: Domain, dtype: object
```

In [23]: `emp['Age'] = emp['Age'].str.replace(r'\W', '', regex=True)`

In [24]: `emp['Age']`

Out[24]:

```
0    34years
1     45yr
2       NaN
3       NaN
4     67yr
5     55yr
Name: Age, dtype: object
```

In [25]: `emp['Age'] = emp['Age'].str.extract('(\d+)') #r'(\d+)' for extracting text`

In [26]: `emp['Age']`

Out[26]:

```
0     34
1     45
2     NaN
3     NaN
4     67
5     55
Name: Age, dtype: object
```

```
In [27]: emp['Location'] = emp['Location'].str.replace(r'\W', '', regex=True)
```

```
In [28]: emp['Location']
```

```
Out[28]: 0      Mumbai
1      Bangalore
2         NaN
3      Hyderabad
4         NaN
5         Delhi
Name: Location, dtype: object
```

```
In [29]: emp
```

```
Out[29]:
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5^00#0	2+
1	Teddy	Testing	45	Bangalore	10%%000	<3
2	Umar	Dataanalyst	NaN	NaN	1\$5%000	4> yrs
3	Jane	Analytics	NaN	Hyderabad	2000^0	NaN
4	Uttam	Statistics	67	NaN	30000-	5+ year
5	Kim	NLP	55	Delhi	6000^\$0	10+

```
In [30]: emp['Salary'] = emp['Salary'].str.replace(r'\W', '', regex=True)
```

```
In [31]: emp['Salary']
```

```
Out[31]: 0      5000
1     10000
2     15000
3     20000
4     30000
5     60000
Name: Salary, dtype: object
```

```
In [32]: emp['Exp'] = emp['Exp'].str.extract('(\d+)') #\d means extracting numbers from
```

```
In [33]: emp['Exp']
```

```
Out[33]: 0      2
1      3
2      4
3     NaN
4      5
5     10
Name: Exp, dtype: object
```

```
In [34]: emp
```

Out[34]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderabad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [35]: `clean_data = emp.copy()`

In [36]: `clean_data`

Out[36]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderabad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

eda techniques

In [37]: `clean_data`

Out[37]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderabad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [38]: `clean_data.isnull().sum()`

```
Out[38]: Name      0
        Domain    0
        Age       2
        Location   2
        Salary     0
        Exp       1
        dtype: int64
```

```
In [39]: clean_data['Age']
```

```
Out[39]: 0      34
        1      45
        2     NaN
        3     NaN
        4      67
        5      55
        Name: Age, dtype: object
```

```
In [40]: import numpy as np
```

```
In [41]: clean_data['Age'] = clean_data['Age'].fillna(np.mean(pd.to_numeric(clean_data['A
```

```
In [42]: clean_data['Age']
```

```
Out[42]: 0      34
        1      45
        2    50.25
        3    50.25
        4      67
        5      55
        Name: Age, dtype: object
```

```
In [43]: clean_data['Exp'] = clean_data['Exp'].fillna(np.mean(pd.to_numeric(clean_data['E
```

```
In [44]: clean_data['Exp']
```

```
Out[44]: 0      2
        1      3
        2      4
        3    4.8
        4      5
        5     10
        Name: Exp, dtype: object
```

```
In [45]: clean_data['Location'].isnull().sum()
```

```
Out[45]: np.int64(2)
```

```
In [46]: clean_data['Location']
```

```
Out[46]: 0      Mumbai
        1    Bangalore
        2         NaN
        3     Hyderbad
        4         NaN
        5        Delhi
        Name: Location, dtype: object
```

```
In [47]: clean_data['Location'] = clean_data['Location'].fillna(clean_data['Location'].mode[0])
```

```
In [48]: clean_data['Location']
```

```
Out[48]: 0      Mumbai
1    Bangalore
2    Bangalore
3     Hyderabad
4    Bangalore
5         Delhi
Name: Location, dtype: object
```

```
In [49]: clean_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Name        6 non-null     object
1   Domain      6 non-null     object
2   Age         6 non-null     object
3   Location    6 non-null     object
4   Salary      6 non-null     object
5   Exp         6 non-null     object
dtypes: object(6)
memory usage: 420.0+ bytes
```

```
In [50]: clean_data['Age'] = clean_data['Age'].astype(int)
```

```
In [51]: clean_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Name        6 non-null     object
1   Domain      6 non-null     object
2   Age         6 non-null     int64
3   Location    6 non-null     object
4   Salary      6 non-null     object
5   Exp         6 non-null     object
dtypes: int64(1), object(5)
memory usage: 420.0+ bytes
```

```
In [52]: clean_data['Salary'] = clean_data['Salary'].astype(int)
```

```
In [53]: clean_data.info()
```



```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Name        6 non-null     object
1   Domain      6 non-null     object
2   Age         6 non-null     int64
3   Location    6 non-null     object
4   Salary      6 non-null     int64
5   Exp         6 non-null     object
dtypes: int64(2), object(4)
memory usage: 420.0+ bytes
```

```
In [54]: clean_data['Exp'] = clean_data['Exp'].astype(int)
```

```
In [55]: clean_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Name        6 non-null     object
1   Domain      6 non-null     object
2   Age         6 non-null     int64
3   Location    6 non-null     object
4   Salary      6 non-null     int64
5   Exp         6 non-null     int64
dtypes: int64(3), object(3)
memory usage: 420.0+ bytes
```

```
In [56]: clean_data['Name'] = clean_data['Name'].astype('category')
clean_data['Domain'] = clean_data['Domain'].astype('category')
clean_data['Location'] = clean_data['Location'].astype('category')
```

```
In [57]: clean_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Name        6 non-null     category
1   Domain      6 non-null     category
2   Age         6 non-null     int64
3   Location    6 non-null     category
4   Salary      6 non-null     int64
5   Exp         6 non-null     int64
dtypes: category(3), int64(3)
memory usage: 938.0 bytes
```

```
In [58]: clean_data
```

Out[58]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50	Bangalore	15000	4
3	Jane	Analytics	50	Hyderbad	20000	4
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [59]: `clean_data.to_csv('clean_data.csv')`

In [60]: `import os`
`os.getcwd() # to get path of the file`

Out[60]: 'C:\\Users\\chamb'

In [61]: `clean_data`

Out[61]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50	Bangalore	15000	4
3	Jane	Analytics	50	Hyderbad	20000	4
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [62]: `import matplotlib.pyplot as plt`
`import seaborn as sns`

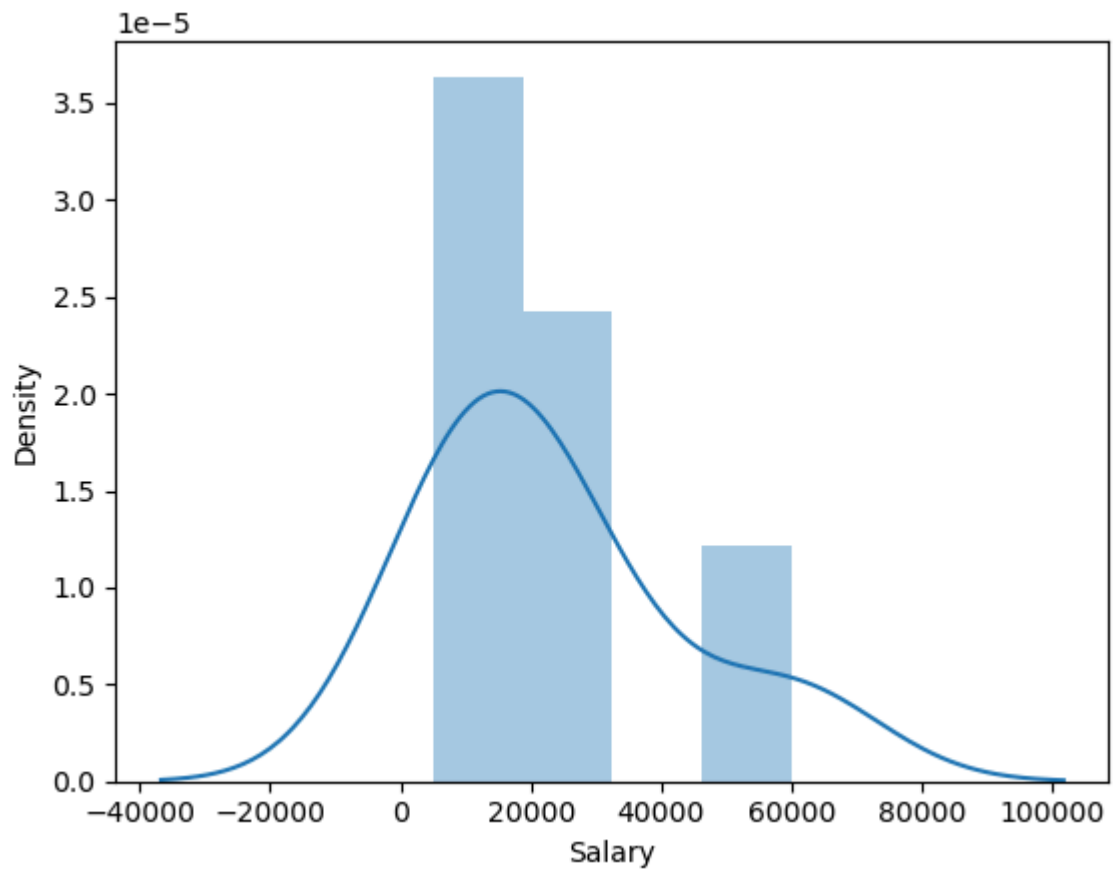
In [63]: `import warnings`
`warnings.filterwarnings('ignore')`

In [64]: `clean_data['Salary']`

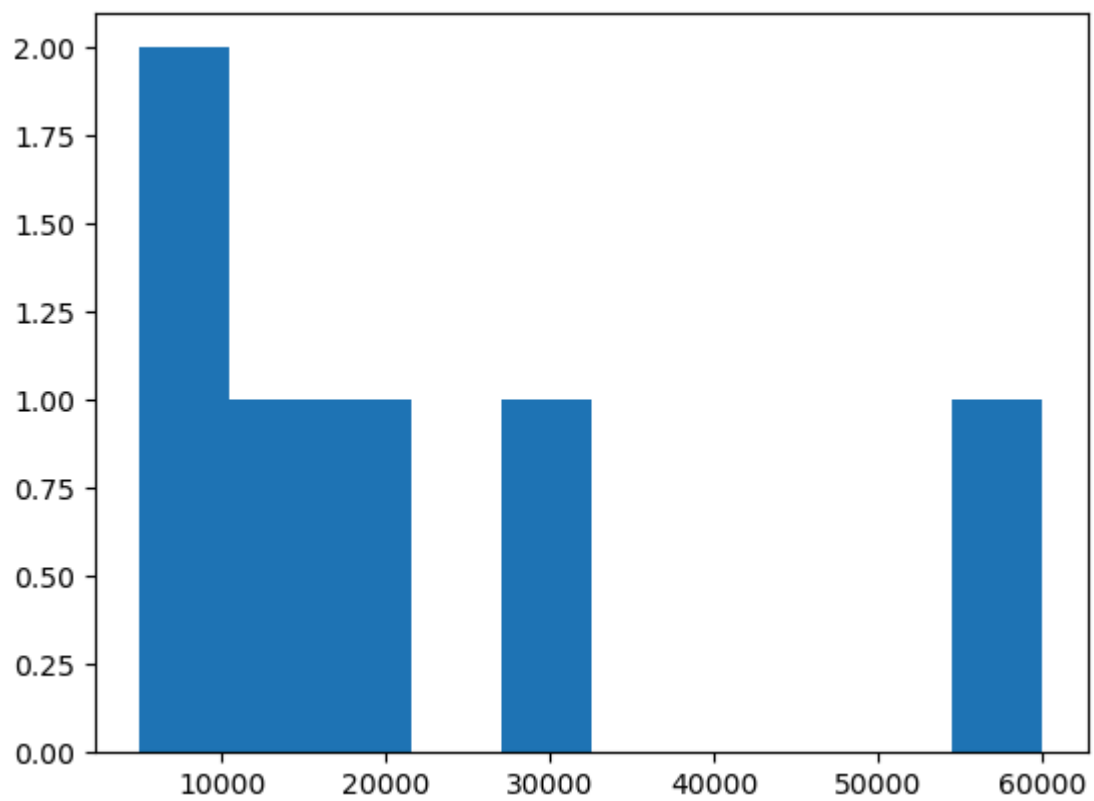
Out[64]:

```
0    5000
1   10000
2   15000
3   20000
4   30000
5   60000
Name: Salary, dtype: int64
```

In [65]: `vis1 = sns.distplot(clean_data['Salary'])`



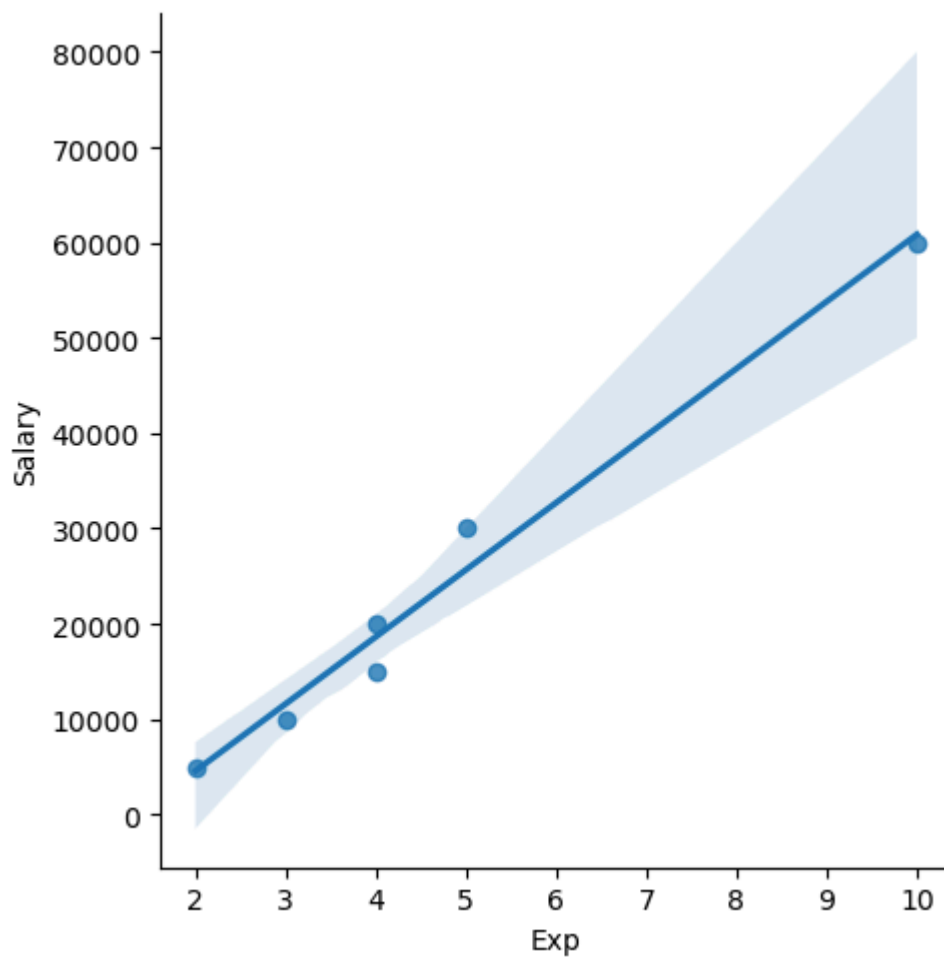
```
In [66]: vis2 = plt.hist(clean_data['Salary'])
```

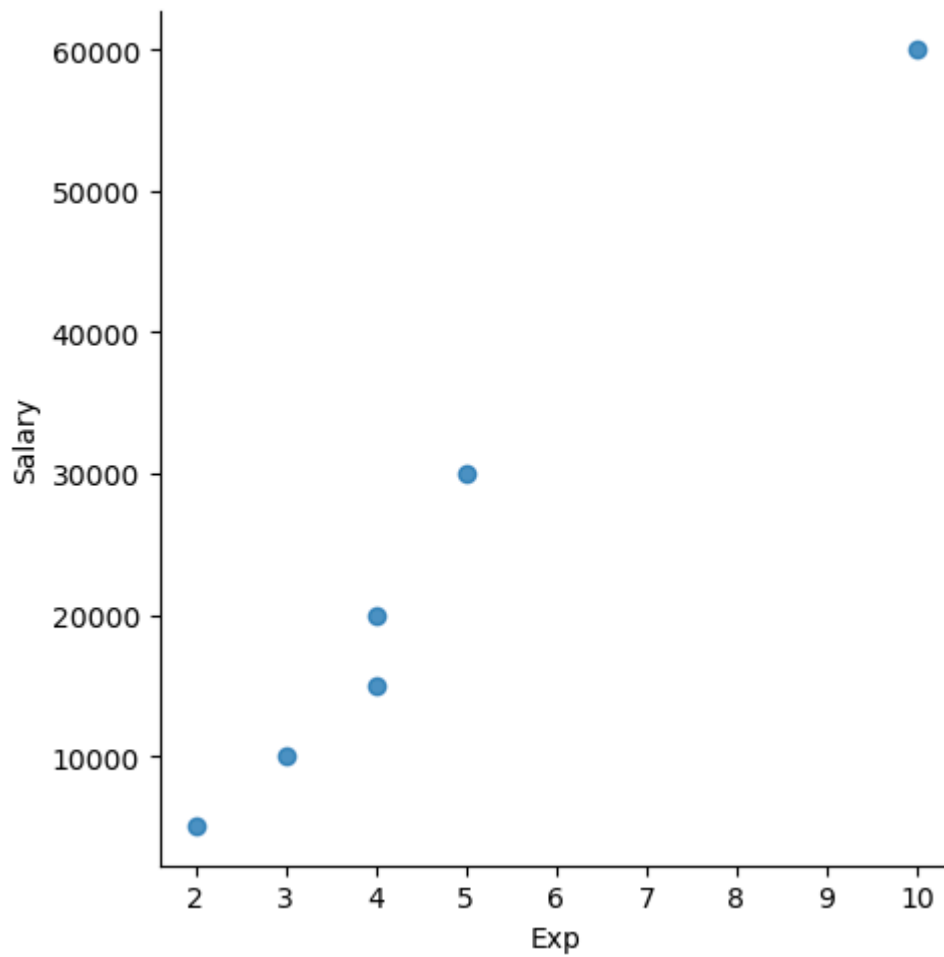


```
In [67]: clean_data
```

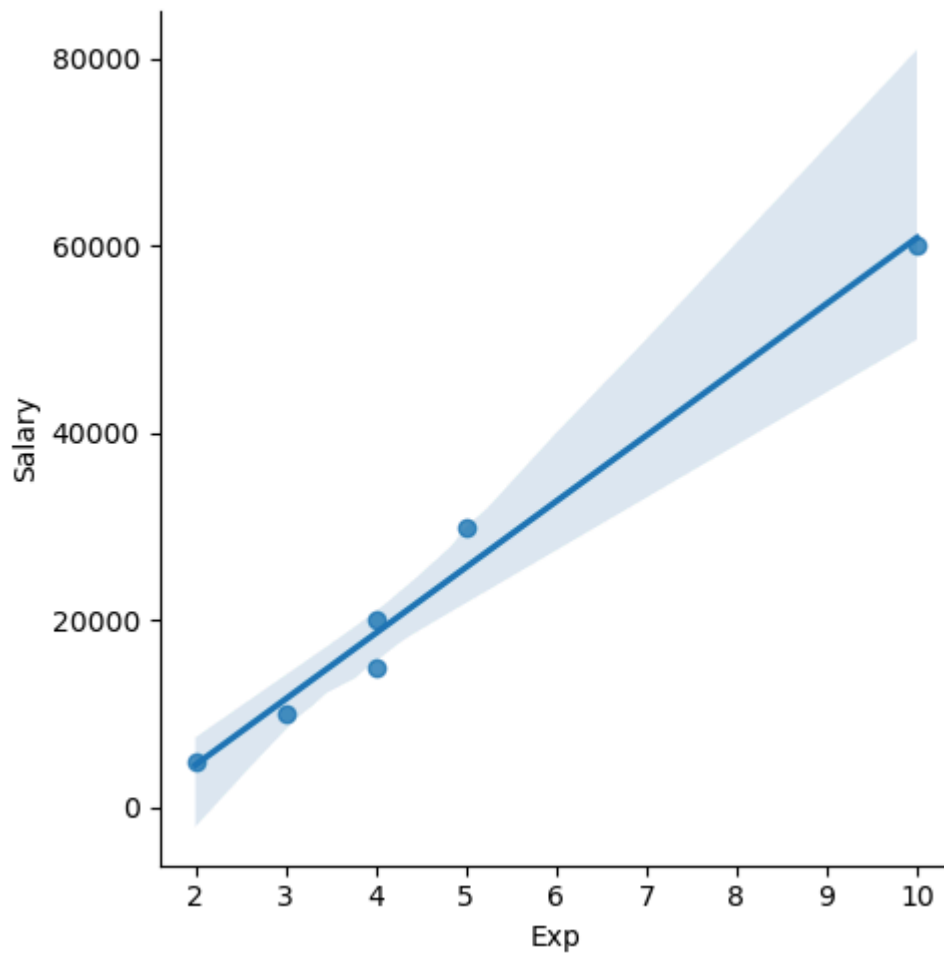
Out[67]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50	Bangalore	15000	4
3	Jane	Analytics	50	Hyderabad	20000	4
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [68]: `vis4 = sns.lmplot(data=clean_data, x = 'Exp', y= 'Salary' )`In [69]: `vis5 = sns.lmplot(data=clean_data, x = 'Exp', y= 'Salary', fit_reg = False)`



```
In [70]: vis5 = sns.lmplot(data=clean_data, x = 'Exp', y= 'Salary', fit_reg = True)
```



```
In [71]: clean_data[:, ]
```

```
Out[71]:
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50	Bangalore	15000	4
3	Jane	Analytics	50	Hyderabad	20000	4
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

```
In [72]: clean_data[:, :-1]
```

Out[72]:

	Name	Domain	Age	Location	Salary	Exp
5	Kim	NLP	55	Delhi	60000	10
4	Uttam	Statistics	67	Bangalore	30000	5
3	Jane	Analytics	50	Hyderabad	20000	4
2	Umar	Dataanalyst	50	Bangalore	15000	4
1	Teddy	Testing	45	Bangalore	10000	3
0	Mike	Datascience	34	Mumbai	5000	2

In [73]: `clean_data.columns`

Out[73]: `Index(['Name', 'Domain', 'Age', 'Location', 'Salary', 'Exp'], dtype='object')`

In [74]: `X_iv = clean_data[['Name', 'Domain', 'Age', 'Location', 'Exp']]`

In [75]: `X_iv`

Out[75]:

	Name	Domain	Age	Location	Exp
0	Mike	Datascience	34	Mumbai	2
1	Teddy	Testing	45	Bangalore	3
2	Umar	Dataanalyst	50	Bangalore	4
3	Jane	Analytics	50	Hyderabad	4
4	Uttam	Statistics	67	Bangalore	5
5	Kim	NLP	55	Delhi	10

In [76]: `y_dv = clean_data[['Salary']]`

In [77]: `y_dv`

Out[77]:

	Salary
0	5000
1	10000
2	15000
3	20000
4	30000
5	60000

In [78]: `emp`

Out[78]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderabad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [79]: `clean_data`

Out[79]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50	Bangalore	15000	4
3	Jane	Analytics	50	Hyderabad	20000	4
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [80]: `imputation = pd.get_dummies(clean_data, dtype=int) #get_dummies convert categori`

In [81]: `imputation`

Out[81]:

	Age	Salary	Exp	Name_Jane	Name_Kim	Name_Mike	Name_Teddy	Name_Umar
0	34	5000	2	0	0	1	0	0
1	45	10000	3	0	0	0	1	0
2	50	15000	4	0	0	0	0	1
3	50	20000	4	1	0	0	0	0
4	67	30000	5	0	0	0	0	0
5	55	60000	10	0	1	0	0	0



In [82]: `clean_data`

Out[82]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50	Bangalore	15000	4
3	Jane	Analytics	50	Hyderbad	20000	4
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [83]: `len(clean_data)`

Out[83]: 6

In [84]: `imputation.columns`

Out[84]: Index(['Age', 'Salary', 'Exp', 'Name_Jane', 'Name_Kim', 'Name_Mike',
 'Name_Teddy', 'Name_Umar', 'Name_Uttam', 'Domain_Analytics',
 'Domain_Dataanalyst', 'Domain_Datascience', 'Domain_NLP',
 'Domain_Statistics', 'Domain_Testing', 'Location_Bangalore',
 'Location_Delhi', 'Location_Hyderabad', 'Location_Mumbai'],
 dtype='object')

In [85]: `len(imputation.columns)`

Out[85]: 19

In []:

In []:

In []:

In []:

In []: